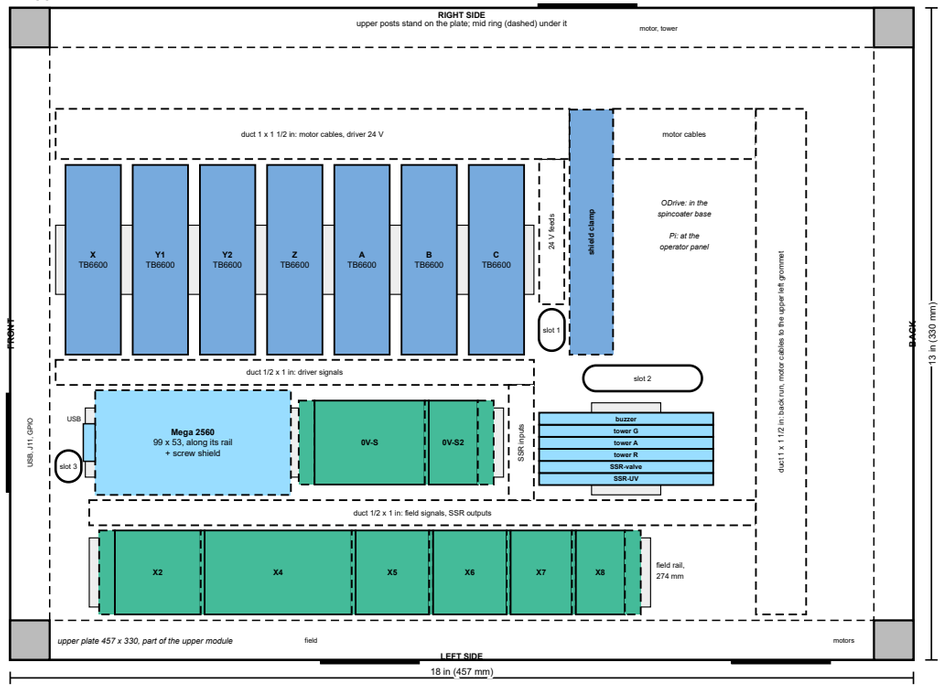
RMR control panel: packing study, 18 x 13 x 11 in

Two decks in a 2020 frame, both on 1/16 in aluminum plates: lower deck for 110 VAC and 24 V power, upper deck for drives and logic; acrylic walls. Scale 1:3.5 when printed at 100 %.

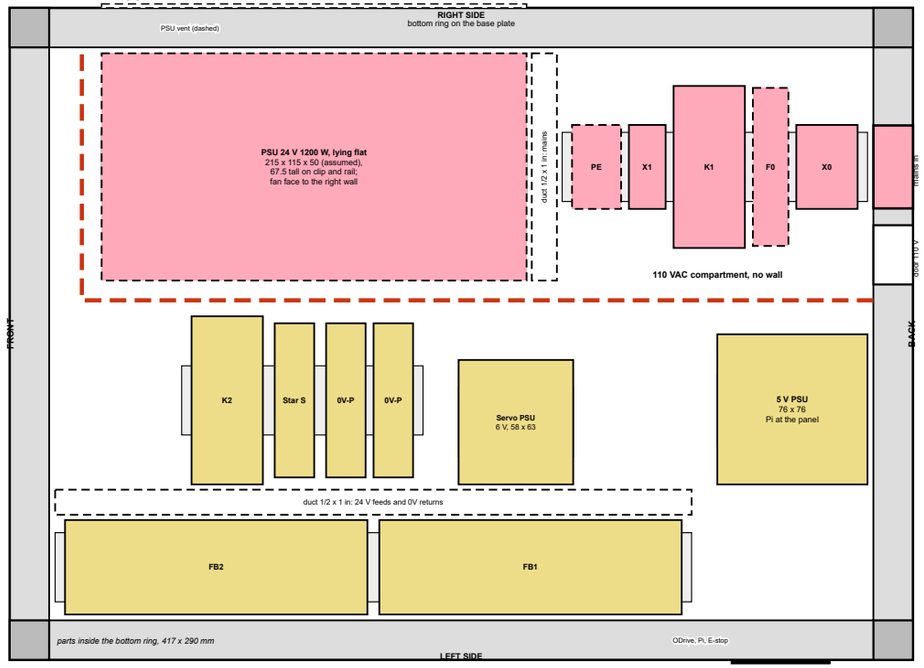
A Upper deck, seen from above

1/16 in aluminum plate on the mid ring; parts at true size



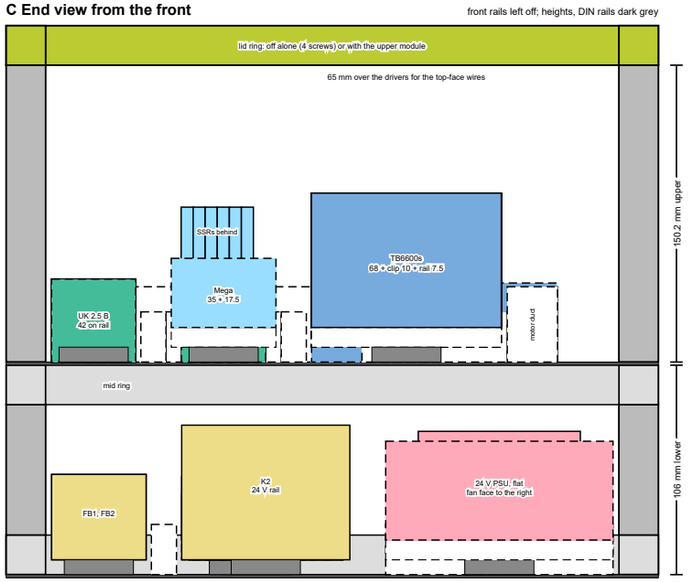
B Lower deck, upper deck lifted off

1/16 in aluminum base plate, bonded to PE



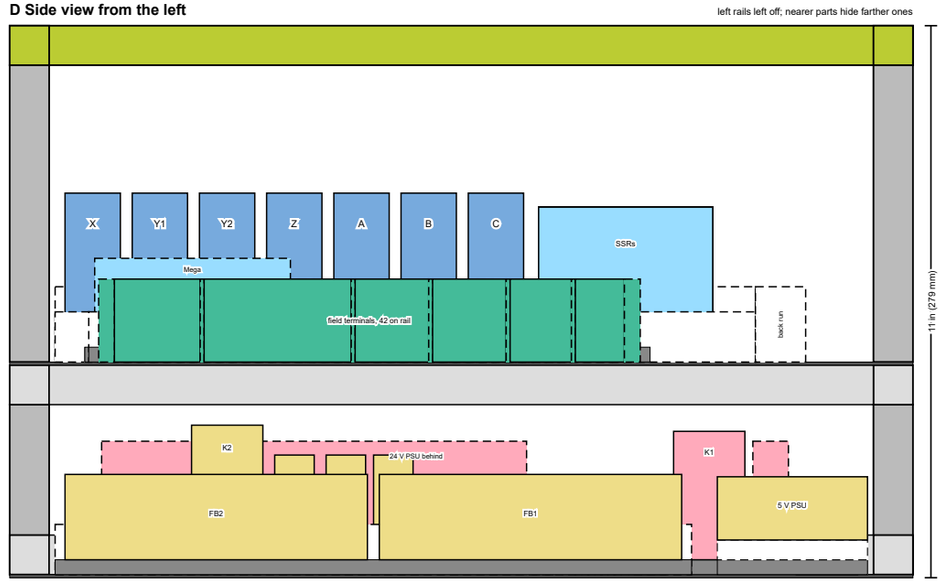
C End view from the front

front rails left off; heights, DIN rails dark grey



D Side view from the left

left rails left off; nearer parts hide farther ones



Zones

- 110 VAC and the PSU
- 24 V distribution and DC-DC
- drives
- logic and SSRs
- field terminals
- 2020 frame (posts darker)
- lid ring, lifts off
- 1/16 in aluminum plate, edge-on
- DIN rail, 7.5 tall (elevations)
- solid outline: measured or datasheet
- dashed: size assumed, measure

Mounting

Every part stands on a 7.5 mm DIN rail; parts without DIN feet add a printed clip, 10 mm. TB6600s stand on their 96 x 28 base, ears 88 apart across the rail, fins vertical. The 24 V PSU lies flat on a 215 x 115 face, its fan face 23 mm from the right panel. The Mega lies along its rail, USB jack to the front grommet; the SSR sockets stand across their own short rail.

Height budget (mm)

base plate 1.6, lower bay 106, upper plate 1.6, rail 7.5 + clip 10 + drivers 68, wire room 64.7, lid ring 20 = 279.4.
The lower left grommet sets the lower bay: its 64 mm ring needs a 66 mm window between the bottom and mid rings. K2 alone would set 78.6, which leaves 38.6 (ADR-0011).

Build and access

Two modules, like a platform-framed house. Lower: base plate, bottom ring and lower posts; stages 3 and 4 run on it alone. Upper: mid ring, upper plate, upper posts and lid ring, built and wired on the bench, then set on 4 studs in the lower posts that locate the decks. Inter-deck wires cross on pluggable terminals. Frame: sheet 4.

Heat and cables

6 mm between drivers for air. The PSU fan blows into the vent in the lower right panel; its slots wait for the fan's size. Five 50 mm grommets: upper left at x 182 (field) and x 390 (motors, at the back run's end), upper front (USB, J11, Pi GPIO), upper right (a motor, the tower), lower left (ODrive 24 V, Pi 5 V, E-stop). Mains in at the back wall in line with the 110 V rail; the 110 V door cable leaves beside it by an M20 gland; no wall around the mains parts (ADR-0013).

Sizes: along x across x high (mm)

TB6600 28 x 96 x 68, measured. From the listings: K1 HC1-63 36 x 82 x 65; K2 BMC7-25U 36 x 85 x 68; FB1 and FB2 153 x 48 x 43; uxcell bar 20 x 78 x 53; SSR 6.1 x 88 x 71 (41F-1Z socket); UK 2.5 B 6.2 x 42.5 x 42; 5 V PSU 76 x 76 x 32; 6 V 63 x 58 x 23; Mega 99 x 53 (A000067). Assumed (dashed): 24 V PSU 215 x 115 x 50, F0, end clamps, PE blocks, shield clamp, Mega height. Heights add the rail.

Layout 3b (ADR-0010)

From the CP-SAT packing study: 36,798 conductor-mm as drawn (35,949 at the study's 78.6 bay), against 57,478 for the hand layout. Each block centre is held to layout3b_build .json; no box overlaps another, and no plate slot lies over the mains (y 182 and up).

Packing study

Sheet 3. Legal, landscape, 100 %. Scale 1:3.5.
Layout 3b (ADR-0010), 2026-10-17: the PSU flat, every part at its real size.
Frame and plates: sheet 4. Measure dashed sizes.